Variant

key	p_a	p_b	p_c	d

Table 1: Theoretical Analysis Results

For each basic-operation set, give the *exact* closed-form count (floors, ceilings and $\log_d$ allowed; no summations), and the asymptotic class. The proofs and the derivation of $A(n)$ (random variables, expectations, treatment of s) go into the corresponding subsections of your report; this table only collects the final results.

Basic operation	Case	Exact count (closed form)	Asymptotic
(1) comparisons	$B_1(n)$		
	$W_1(n)$		
	$A_1(n)$		
(2) loop increments	$B_2(n)$		
	$W_2(n)$		
	$A_2(n)$		
(3) assignments	$B_3(n)$		
	$W_3(n)$		
	$A_3(n)$		
(4) assignments	$B_4(n)$		
	$W_4(n)$		
	$A_4(n)$		

Table 2: Real Execution Times

Basic operation identified as the "correct" one: _____

Programming language / machine: _____

n	**Best (s)**	**Worst (s)**	**Average (s)**
1			
5			
10			
20			
40			
50			
60			
80			
100			
150			
200			
250			
300			
350			
400			

Table 3: Exact Count Verification (Best and Worst Case)

For each case (1)–(4) and each n below, run your algorithm on your best-case and worst-case inputs and enter the counted values together with the values given by your $B(n)$ and $W(n)$ formulas. They are expected to be equal. If a counted value does not match your formula, report it anyway and explain the reason in your report.

Basic operation	n	**Counted best**	**Formula** $B(n)$	**Counted worst**	**Formula** $W(n)$
(1) comparisons	1				
	5				
	10				
	20				
	40				
	100				
(2) loop increments	1				
	5				
	10				
	20				
	40				
	100				
(3) assignments	1				
	5				
	10				
	20				
	40				
	100				
(4) assignments	1				
	5				
	10				
	20				
	40				
	100				

Table 4: Exact Count Verification (Average Case)

For each case (1)–(4) and each n below, run your algorithm on at least 1000 random inputs generated with the probabilities of your variant, and enter the mean of the counted values together with the value given by your $A(n)$ formula. Comment on the difference in your report.
Number of random inputs used for each n ($N \geq 1000$): _____

Basic operation	n	**Mean of counted values**	**Formula** $A(n)$	**Difference**
(1) comparisons	10			
	40			
	100			
(2) loop increments	10			
	40			
	100			
(3) assignments	10			
	40			
	100			
(4) assignments	10			
	40			
	100			